

Section C — Mini Capstone Project

1. Three Hardware Components to Check First When a Newly Assembled PC Fails to Boot

When a freshly assembled desktop fails to power on or halts before reaching the operating system, check these three components first in strict order:

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1. Power Supply Unit (PSU) & Power Connections:

- **Why:** Without correct, stable DC voltages (+12V, +5V, +3.3V), no other hardware can function. In newly assembled systems, human error during cabling is the most common cause of complete power failure (such as leaving the main 24-pin ATX power cable loosely latched, forgetting the dedicated 8-pin EPS CPU power connector at the top left of the motherboard, or flipping the rear rocker switch to 0 instead of I).

2. RAM (System Memory):

- **Why:** Unseated or partially locked memory modules are the single most frequent cause of a system turning on (fans spinning) but failing the **Power-On Self-Test (POST)** with a blank display and motherboard debug LEDs stopping on DRAM. The memory controller cannot complete initial handshakes unless both retention clips are locked and contacts make firm physical connection.

3. CPU Installation & Cooler Mounting:

- **Why:** If the CPU is misaligned, has bent socket pins, or if the thermal cooler baseplate is mounted with uneven pressure, the motherboard's power protection circuitry halts the boot sequence immediately to prevent catastrophic electrical or thermal damage. Furthermore, if the cooler's fan is plugged into a chassis fan header instead of CPU_FAN, the BIOS may halt with a CPU Fan Error.

2. How Motherboard–CPU Compatibility Affects System Functionality

- **Reference System:** AMD Ryzen 5 7600X CPU paired with an ASUS Prime B650M-A WiFi Motherboard.
- **Physical Socket Compatibility:** The processor and motherboard must share the exact physical socket architecture. The Ryzen 5 7600X requires an **AM5 (LGA 1718)** socket. An incompatible socket (such as older AM4 PGA or Intel LGA 1700) will physically not fit, and forcing it will crush or snap fragile socket pins.
- **Chipset & Power Delivery (VRM):** The AMD B650 chipset provides the necessary PCIe 4.0/5.0 communication lanes and dedicated voltage regulation modules (VRMs) required to deliver stable voltage to a 105WTDP processor under load.
- **BIOS / Firmware Revision Impact:** Even when physical socket dimensions match, a motherboard manufactured before a specific processor's launch may contain outdated UEFI/BIOS firmware that lacks the microcode necessary to detect and initialize the CPU. If the BIOS version does not support the processor out of the box, the system will power on, but the screen will stay completely black, and the motherboard debug LED will remain stuck on CPU.

3. Verification of RAM and Storage Device Installation

Physical Inspection

- **RAM:** Check that the notch on the memory stick is aligned with the key in the DIMM slot and that modules are installed in the recommended dual-channel slots (Slots **A2** and **B2** / 2nd and 4th slots from the CPU). Confirm that both outer retaining clips are snapped fully into the locked position.
- **Storage (M.2 NVMe SSD):** Ensure the M.2 SSD is fully inserted into the M-key slot at a 30-degree angle, pressed flat against the standoff, and secured firmly with the mounting screw or toolless latch. For 2.5-inch SATA drives, confirm that both the SATA data cable (connected to the motherboard) and SATA power cable (from the PSU) are clicked in securely.

BIOS/POST Confirmation

Enter the UEFI/BIOS Setup Utility by pressing **DEL** or **F2** upon power-up and confirm the following:

- **Total Installed Memory:** The BIOS main screen must report the full expected capacity (e.g., **16,384 MB / 16 GB**) and show **Dual-Channel Mode** active at rated frequency (e.g., DDR5-5200 MT/s). If only 8 GB is reported, one module is unseated or defective.
- **Storage Device Detection:** Under the **Storage Information / NVMe Configuration** section, the drive's exact hardware string (e.g., Crucial P3 Plus 1TB or Samsung SSD 980) must appear with its full storage capacity.
- **Boot Priority:** Under the **Boot** menu, the drive (or Windows Boot Manager) must be visible and selectable as **Boot Option #1** to confirm the UEFI bootloader is ready for OS deployment.

4. Hardware-Installation Checklist to Prevent Deployment Failures

Step	Installation Phase	Verification Action
1	Pre-Mounting & ESD Protection	Work on a clean, grounded surface using an anti-static wrist strap. Pre-install the CPU, M.2 NVMe SSD, RAM, and CPU cooler onto the motherboard outside the case on top of the motherboard box before lowering it into the chassis. PNG
2	Motherboard Standoff Verification	Verify that brass motherboard standoffs inside the case match the motherboard's mounting holes only. Remove any extra standoffs behind the board to prevent direct electrical short circuits against the metal tray.
3	Dual-Check Power Cable Latching	Check that the 24-pin ATX main power cable and the 8-pin (4+4) EPS CPU power cable are plugged in completely until their plastic retaining clips snap firmly into place. Ensure dedicated PCIe cables are used for discrete graphics cards. PNG
4	Firmware & Single-Stick Validation	Before mass casing, perform a test power-on bench test to confirm the board displays video and passes POST on a single stick of RAM, updating the motherboard BIOS to the latest stable release via USB BIOS FlashBack if required.

5. Technical Justification: SSDs vs. HDDs for ABC Solutions Computer Lab

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For the ABC Solutions workstation lab, equipping systems with **M.2 NVMe Solid-State Drives (SSDs)** instead of traditional mechanical **Hard Disk Drives (HDDs)** is essential for the following technical reasons:

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- **Boot & Application Loading Speeds:** NVMe SSDs deliver sequential read speeds exceeding 3,500 to 5,000 MB/s, compared to just 100–150 MB/s on a 7200 RPM HDD. Workstations boot into Windows in under 10 seconds, allowing classes and lab sessions to begin immediately without lost instructional time.
- **Concurrent Multi-User Responsiveness (Random IOPS):** In a computer lab, students frequently launch multiple productivity applications, switch between heavy browser tabs, and compile code simultaneously. NVMe SSDs provide over 400,000 IOPS (Input/Output Operations Per Second) with sub-millisecond seek latency (< 0.05 ms vs. 10–15 ms on mechanical platters), eliminating system lag and preventing the "100% Disk Usage" freezes common on HDDs.
- **Durability & Zero Acoustic Noise:** HDDs rely on high-speed spinning platters and fragile moving actuator heads that are susceptible to physical shock, wear, and mechanical failure when desks are bumped in a shared lab. SSDs use non-volatile NAND flash with no moving parts, providing silent operation and higher long-term reliability.